

A Rigorous Finite-Level KMS Theorem Behind Lemma 1

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Abstract

We give a self-contained rigorous version of the finite-level KMS statement used in Lemma 1 of the note *Finite-Level S -Adele Models for the Bost-Connes System*. The exact statement in that note writes a crossed product

$$B_S = C_c^\infty(X_S) \rtimes \mathbb{R}_+^\times$$

with a continuous scaling group, but its proof sketch uses the discrete semigroup generated by the primes in S . These are not literally the same object. The theorem proved here isolates the precise finite-prime Bost-Connes type system for which the KMS statement is rigorous. Its unique Gibbs–KMS state has partition function

$$Z_S^{\text{fin}}(\beta) = \prod_{p \in S_f} (1 - p^{-\beta})^{-1}.$$

The archimedean factor

$$Z_\infty(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right)$$

is then obtained as a genuine Mellin-Gaussian normalization. Thus the full expression

$$Z_S(\beta) = Z_S^{\text{fin}}(\beta) Z_\infty(\beta)$$

is proved rigorously as the product of the finite-prime KMS partition function and the archimedean Mellin factor. We also record exactly what additional identification would be needed to turn this into a proof for the formal crossed product on X_S itself.

1 The statement to be made precise

Let $S \supset \{\infty\}$ be a finite set of places of $\mathbb{Q}$, and write

$$S_f := S \setminus \{\infty\}$$

the finite set of rational primes belonging to S . The lemma in the original note asserts, in effect, that for $\Re(\beta) > 1$ there is a unique KMS_β state with partition function

$$Z_S(\beta) = \left(\prod_{p \in S_f} (1 - p^{-\beta})^{-1} \right) \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right). \quad (1)$$

There are two important points to separate.

- (i) A KMS_β state is normally defined for a real inverse temperature β . The partition function may then admit holomorphic continuation to a complex half-plane, but the KMS condition itself is a condition at real β .

- (ii) The finite Euler product in (1) comes naturally from the discrete semigroup generated by the finite primes in S , not from the continuous group $\mathbb{R}_+^\times$ by itself.

Accordingly, the rigorous theorem below uses the finite-prime semigroup

$$\Lambda_S := \left\{ \prod_{p \in S_f} p^{k_p} : k_p \in \mathbb{N} \right\}, \quad (2)$$

where $\mathbb{N} = \{0, 1, 2, \dots\}$. Thus $1 \in \Lambda_S$. If $S_f = \emptyset$, then $\Lambda_S = \{1\}$.

2 The finite-prime Toeplitz system

Let

$$\mathcal{H}_S := \ell^2(\Lambda_S),$$

with orthonormal basis $\{e_n : n \in \Lambda_S\}$. For $m \in \Lambda_S$, define an isometry

$$V_m e_n := e_{mn}.$$

Then

$$V_m V_n = V_{mn}, \quad V_1 = 1.$$

Let

$$\mathcal{T}_S := C^*(1, V_p : p \in S_f) \subset B(\mathcal{H}_S),$$

the unital C^* -algebra generated by the semigroup isometries. Equivalently, $\mathcal{T}_S$ is the Toeplitz algebra of the free abelian semigroup $\Lambda_S \cong \mathbb{N}^{|S_f|}$. With this convention, if $S_f = \emptyset$, then $\mathcal{H}_S = \mathbb{C}$ and $\mathcal{T}_S = \mathbb{C}$. The linear span of monomials

$$V_m V_n^*, \quad m, n \in \Lambda_S,$$

is a dense $*$ -subalgebra of $\mathcal{T}_S$.

Define a self-adjoint diagonal operator H_S on $\mathcal{H}_S$ by

$$H_S e_n = (\log n) e_n.$$

Since $n \geq 1$, this is a positive self-adjoint operator. It implements the time evolution

$$\alpha_t(a) := e^{itH_S} a e^{-itH_S}, \quad a \in \mathcal{T}_S.$$

On generators,

$$\alpha_t(V_m) = m^{it} V_m. \quad (3)$$

This is the finite-prime version of the Bost-Connes time evolution.

3 The finite Euler product as a trace partition function

Lemma 1. *For every real $\beta > 0$, the operator $e^{-\beta H_S}$ is trace class and*

$$Z_S^{\text{fin}}(\beta) := \text{Tr}(e^{-\beta H_S}) = \sum_{n \in \Lambda_S} n^{-\beta} = \prod_{p \in S_f} (1 - p^{-\beta})^{-1}. \quad (4)$$

In particular, the formula holds for every real $\beta > 1$. Moreover, the right-hand side is holomorphic for $\Re(\beta) > 0$.

Proof. Since $H_S e_n = (\log n) e_n$, we have

$$e^{-\beta H_S} e_n = n^{-\beta} e_n.$$

Therefore

$$\mathrm{Tr}(e^{-\beta H_S}) = \sum_{n \in \Lambda_S} n^{-\beta}.$$

By unique factorization inside the finite semigroup Λ_S , every $n \in \Lambda_S$ has a unique expression

$$n = \prod_{p \in S_f} p^{k_p}, \quad k_p \in \mathbb{N}.$$

Hence, for $\beta > 0$,

$$\sum_{n \in \Lambda_S} n^{-\beta} = \sum_{(k_p) \in \prod_{p \in S_f} \mathbb{N}} \prod_{p \in S_f} p^{-\beta k_p} = \prod_{p \in S_f} \sum_{k=0}^{\infty} p^{-\beta k} = \prod_{p \in S_f} (1 - p^{-\beta})^{-1}.$$

The finite product is holomorphic wherever each geometric series converges, namely for $\Re(\beta) > 0$. If $S_f \neq \emptyset$, the endpoint $\beta = 0$ is sharp: then $n^{-\beta} = 1$ for every $n \in \Lambda_S$, and the infinite sum diverges. If $S_f = \emptyset$, then $\Lambda_S = \{1\}$ and the partition function is identically 1. This proves the claim. $\square$

4 Existence and uniqueness of the finite-prime KMS state

For $\beta > 0$, define

$$\varphi_{\beta,S}(a) := \frac{\mathrm{Tr}(e^{-\beta H_S} a)}{Z_S^{\mathrm{fin}}(\beta)}, \quad a \in \mathcal{T}_S. \quad (5)$$

This is a state because $e^{-\beta H_S}$ is positive trace class and $Z_S^{\mathrm{fin}}(\beta) = \mathrm{Tr}(e^{-\beta H_S})$.

Theorem 1. *For every real $\beta > 0$, the state $\varphi_{\beta,S}$ is the unique KMS_β state on the finite-prime Toeplitz system $(\mathcal{T}_S, \alpha)$. On the dense $*$ -subalgebra spanned by $V_m V_n^*$, it is given by*

$$\varphi_{\beta,S}(V_m V_n^*) = \begin{cases} m^{-\beta}, & m = n, \\ 0, & m \neq n. \end{cases} \quad (6)$$

Proof. First we verify that $\varphi_{\beta,S}$ has the values claimed in (6). For $k \in \Lambda_S$,

$$\langle e_k, V_m V_n^* e_k \rangle$$

can be nonzero only if n divides k inside Λ_S and $mk/n = k$. The latter equality forces $m = n$. Therefore, if $m \neq n$,

$$\mathrm{Tr}(e^{-\beta H_S} V_m V_n^*) = 0.$$

If $m = n$, then $V_m V_m^*$ is the projection onto the closed span of $\{e_{mr} : r \in \Lambda_S\}$. Hence

$$\mathrm{Tr}(e^{-\beta H_S} V_m V_m^*) = \sum_{r \in \Lambda_S} (mr)^{-\beta} = m^{-\beta} Z_S^{\mathrm{fin}}(\beta).$$

Dividing by $Z_S^{\mathrm{fin}}(\beta)$ gives (6).

The Gibbs state (5) satisfies the KMS_β condition. To see this directly on the dense analytic $*$ -subalgebra spanned by the monomials, let

$$A = V_m V_n^*.$$

Then A is entire analytic and

$$\alpha_{i\beta}(A) = (m/n)^{-\beta} A.$$

Writing $\rho_\beta := e^{-\beta H_S}$, the implementation of the dynamics gives

$$\rho_\beta A = \alpha_{i\beta}(A) \rho_\beta.$$

Hence, for analytic monomials A, B , the operator $\rho_\beta A$ is trace class and

$$\text{Tr}(\rho_\beta AB) = \text{Tr}(B \rho_\beta A) = \text{Tr}(B \alpha_{i\beta}(A) \rho_\beta) = \text{Tr}(\rho_\beta B \alpha_{i\beta}(A)).$$

After division by $Z_S^{\text{fin}}(\beta)$, this is precisely

$$\varphi_{\beta,S}(AB) = \varphi_{\beta,S}(B \alpha_{i\beta}(A)).$$

By linearity the same identity holds on the analytic $*$ -algebra spanned by the monomials, and the usual density argument gives the KMS_β condition on $\mathcal{T}_S$.

It remains to prove uniqueness. Let ψ be any KMS_β state on $(\mathcal{T}_S, \alpha)$. Every KMS state is invariant under the dynamics, so

$$\psi(\alpha_t(V_m V_n^*)) = \psi(V_m V_n^*)$$

for all $t \in \mathbb{R}$. But by (3),

$$\alpha_t(V_m V_n^*) = (m/n)^{it} V_m V_n^*.$$

If $m \neq n$, then there exists t such that $(m/n)^{it} \neq 1$, and hence

$$\psi(V_m V_n^*) = 0.$$

If $m = n$, use the KMS condition with $a = V_m$ and $b = V_m^*$. Since

$$\alpha_{i\beta}(V_m) = m^{-\beta} V_m,$$

we obtain

$$\psi(V_m V_m^*) = \psi(V_m^* \alpha_{i\beta}(V_m)) = m^{-\beta} \psi(V_m^* V_m) = m^{-\beta} \psi(1) = m^{-\beta}.$$

Thus ψ agrees with $\varphi_{\beta,S}$ on every monomial $V_m V_n^*$. Since their linear span is dense in $\mathcal{T}_S$, $\psi = \varphi_{\beta,S}$. This proves uniqueness. $\square$

5 The archimedean Gamma factor

The finite-prime KMS theorem accounts rigorously for the finite Euler product. The factor

$$\pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right)$$

is an archimedean Mellin factor. It is obtained from the Gaussian at the real place.

Let

$$d\mu_\infty(x) := 2e^{-\pi x^2} \frac{dx}{x}, \quad x \in (0, \infty).$$

For real $\beta > 0$, define

$$Z_\infty(\beta) := \int_0^\infty x^\beta d\mu_\infty(x) = 2 \int_0^\infty x^{\beta-1} e^{-\pi x^2} dx. \quad (7)$$

Lemma 2. For $\Re(\beta) > 0$,

$$Z_\infty(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right). \quad (8)$$

In particular, (8) holds for every real $\beta > 1$.

Proof. Using the substitution $u = \pi x^2$, so that

$$x = (u/\pi)^{1/2}, \quad \frac{dx}{x} = \frac{1}{2} \frac{du}{u},$$

we compute

$$2 \int_0^\infty x^\beta e^{-\pi x^2} \frac{dx}{x} = 2 \int_0^\infty \left(\frac{u}{\pi}\right)^{\beta/2} e^{-u} \frac{1}{2} \frac{du}{u} = \pi^{-\beta/2} \int_0^\infty u^{\beta/2-1} e^{-u} du.$$

The last integral is $\Gamma(\beta/2)$, valid for $\Re(\beta) > 0$. This proves the formula. $\square$

The corresponding normalized archimedean probability measure is

$$d\nu_{\beta,\infty}(x) := \frac{x^\beta}{Z_\infty(\beta)} d\mu_\infty(x) = \frac{2x^{\beta-1}e^{-\pi x^2}}{\pi^{-\beta/2}\Gamma(\beta/2)} dx. \quad (9)$$

This is a genuine Radon probability measure on $(0, \infty)$ for every real $\beta > 0$.

6 The combined finite-level state and partition function

Define the product algebra using the minimal (spatial) tensor product,

$$\mathcal{A}_S := \mathcal{T}_S \otimes_{\min} C_0((0, \infty)),$$

with dynamics

$$\sigma_t := \alpha_t \otimes \text{id}.$$

The algebra $C_0((0, \infty))$, and hence $\mathcal{A}_S$, is non-unital but σ -unital. We use the standard non-unital formulation of the KMS condition: the KMS identity is checked on a dense algebra of entire analytic elements and then extended to the C^* -algebra by continuity, equivalently by applying the usual approximate-unit argument; see, for example, [2].

For real $\beta > 0$, define a state on elementary tensors by

$$\omega_{\beta,S}(a \otimes f) := \varphi_{\beta,S}(a) \int_0^\infty f(x) d\nu_{\beta,\infty}(x), \quad (10)$$

and extend linearly and continuously to $\mathcal{A}_S$.

Proposition 1. For every real $\beta > 0$, the state $\omega_{\beta,S}$ is a KMS_β state for $(\mathcal{A}_S, \sigma)$. More generally, every KMS_β state on $(\mathcal{A}_S, \sigma)$ is of the form $\varphi_{\beta,S} \otimes \rho$, where ρ is a state on $C_0((0, \infty))$. Consequently, among KMS states whose archimedean marginal is the probability measure $\nu_{\beta,\infty}$, $\omega_{\beta,S}$ is unique. Its normalizing factor is

$$Z_S(\beta) = Z_S^{\text{fin}}(\beta) Z_\infty(\beta) = \left(\prod_{p \in S_f} (1 - p^{-\beta})^{-1} \right) \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right). \quad (11)$$

Proof. The state $\varphi_{\beta,S}$ is KMS_β for $(\mathcal{T}_S, \alpha)$ by Theorem 1. The state $f \mapsto \int f d\nu_{\beta,\infty}$ is invariant under the trivial dynamics on $C_0((0, \infty))$. Therefore their tensor product is KMS_β for $\alpha \otimes \text{id}$. Explicitly, for analytic elementary tensors,

$$\omega_{\beta,S}((a \otimes f)\sigma_{i\beta}(b \otimes g)) = \varphi_{\beta,S}(a\alpha_{i\beta}(b)) \int fg d\nu_{\beta,\infty},$$

while

$$\omega_{\beta,S}((b \otimes g)(a \otimes f)) = \varphi_{\beta,S}(ba) \int gf d\nu_{\beta,\infty}.$$

The two quantities are equal by the finite-prime KMS condition and commutativity of the archimedean factor. Density gives the result for $\mathcal{A}_S$.

The formula for $Z_S(\beta)$ follows immediately by multiplying Lemma 1 and Lemma 2.

It remains to justify the uniqueness statement in the non-unital tensor product. Let ω be any KMS_β state on $(\mathcal{A}_S, \sigma)$. For $0 \leq f \in C_0((0, \infty))$, define a positive functional on $\mathcal{T}_S$ by

$$\psi_f(a) := \omega(a \otimes f).$$

If $\psi_f(1) > 0$, then the normalized functional $\psi_f/\psi_f(1)$ is a KMS_β state on $(\mathcal{T}_S, \alpha)$. Indeed, applying the KMS identity in $\mathcal{A}_S$ to $a \otimes f^{1/2}$ and $b \otimes f^{1/2}$ gives

$$\psi_f(a\alpha_{i\beta}(b)) = \psi_f(ba)$$

for analytic $a, b \in \mathcal{T}_S$. By Theorem 1, $\psi_f = \psi_f(1)\varphi_{\beta,S}$. If $\psi_f(1) = 0$, positivity gives $\psi_f = 0$, so the same conclusion holds.

Now set

$$\rho(f) := \omega(1 \otimes f), \quad f \in C_0((0, \infty)).$$

Then ρ is a positive functional. If (u_j) is a positive contractive approximate unit for $C_0((0, \infty))$, then $1 \otimes u_j$ is an approximate unit for $\mathcal{A}_S$, so $\rho(u_j) \rightarrow \|\omega\| = 1$. Hence ρ is a state. The preceding paragraph shows

$$\omega(a \otimes f) = \varphi_{\beta,S}(a)\rho(f)$$

for all $a \in \mathcal{T}_S$ and all positive $f \in C_0((0, \infty))$, and therefore for all f by linearity. Thus $\omega = \varphi_{\beta,S} \otimes \rho$. Taking $\rho(f) = \int f d\nu_{\beta,\infty}$ gives the particular state (10), and it is the only KMS state with that archimedean marginal. $\square$

Remark 1 (Can the archimedean factor be made operator-algebraic?). *Yes, but not in quite the same trace-class way as the finite-prime Toeplitz factor. Let*

$$\Phi_\infty(f) := 2 \int_0^\infty f(x) e^{-\pi x^2} \frac{dx}{x}$$

be the natural Gaussian Mellin weight on $C_0((0, \infty))$. If M_x denotes multiplication by x , then formally

$$Z_\infty(\beta) = \Phi_\infty(M_x^\beta)$$

for $\beta > 0$. This is a legitimate commutative operator-algebraic normalization by a semifinite weight, but it is not a discrete trace partition function $\text{Tr}(e^{-\beta H})$ on ℓ^2 unless one adds an additional discretization. In particular, nothing in the present construction identifies the product normalization $Z_S^{\text{fin}}(\beta)Z_\infty(\beta)$ with $\text{Tr}(e^{-\beta H})$ for a single trace-class heat operator on a Hilbert-space representation of $\mathcal{A}_S$. For this reason the paper keeps the finite-prime trace-class Gibbs normalization and the archimedean Mellin normalization conceptually separate.

7 Conclusion and relation to the original Lemma 1

We have proved the following rigorous finite-level replacement for Lemma 1.

Theorem 2 (Rigorous finite-level partition theorem). *Let $S \supset \{\infty\}$ be finite and let $S_f = S \setminus \{\infty\}$. For every real $\beta > 0$, the finite-prime Toeplitz system $(\mathcal{T}_S, \alpha)$ has a unique KMS_β state. Its finite-prime partition function is*

$$Z_S^{\text{fin}}(\beta) = \prod_{p \in S_f} (1 - p^{-\beta})^{-1}.$$

The archimedean Gaussian Mellin normalization is

$$Z_\infty(\beta) = \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right).$$

Consequently the combined finite-level normalization is

$$Z_S(\beta) = \left(\prod_{p \in S_f} (1 - p^{-\beta})^{-1} \right) \pi^{-\beta/2} \Gamma\left(\frac{\beta}{2}\right).$$

For each fixed finite S , the same formula is holomorphic throughout the half-plane $\Re(\beta) > 0$. The smaller half-plane $\Re(\beta) > 1$ becomes natural only when one lets S increase through all primes and asks for convergence to the full Euler product.

Remark 2 (What this proves, and what it does not prove). *The theorem above proves the partition function and the finite-prime KMS uniqueness statement rigorously. It does not, by itself, prove that the formal crossed product*

$$C_c^\infty(X_S) \rtimes \mathbb{R}_+^\times$$

has a unique KMS_β state with the same partition function. To prove that stronger statement one would still need an explicit theorem identifying the relevant X_S -crossed-product dynamics with the finite-prime Toeplitz dynamics above, together with a proof that the archimedean Gamma factor arises from the same KMS structure rather than from a separate Mellin normalization. Without those additional arguments, the honest rigorous statement is the finite-prime KMS theorem plus the archimedean Mellin factor proved here.

Remark 3 (Why the finite-level convergence domain is larger). *At fixed finite S , the finite Euler product converges for $\Re(\beta) > 0$, not merely for $\Re(\beta) > 1$. The half-plane $\Re(\beta) > 1$ becomes natural when one lets S increase through all primes and asks for convergence to the full Euler product $\zeta(\beta)$.*

References

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